WORKED SOLUTION

STEP I 1997, Question 7

Pure • Gaussian Quadrature

STEP I 1997, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q7

The key idea

Both sides depend *linearly* on P, so the formula holds for every cubic as soon as it holds for the four basic cubics 1, t, t^2 and t^3. That turns the problem into four equations in four unknowns. They are nonlinear, but the odd and even powers separate neatly, and symmetry about $t = 0$ does the rest.

Solution

Write $\mathrm{P}(t) = c_0 + c_1 t + c_2 t^2 + c_3 t^3$. Then

$$\int_{-1}^{1} \mathrm{P}(t)\,\mathrm{d}t = c_0 \int_{-1}^{1} 1\,\mathrm{d}t + c_1 \int_{-1}^{1} t\,\mathrm{d}t + c_2 \int_{-1}^{1} t^2\,\mathrm{d}t + c_3 \int_{-1}^{1} t^3\,\mathrm{d}t,$$

and $a_1\mathrm{P}(u_1) + a_2\mathrm{P}(u_2)$ splits in exactly the same way. So the formula holds for all cubics if and only if it holds for $\mathrm{P}(t) = t^k$, $k = 0, 1, 2, 3$. ("Only if" because each t^k is itself a cubic polynomial in the broad sense; "if" by adding up.) Since $\int_{-1}^{1} t^k\,\mathrm{d}t$ is 2, 0, $\frac{2}{3}$, 0 for $k = 0, 1, 2, 3$, we need

$$a_1 + a_2 = 2, \tag{A}$$
$$a_1 u_1 + a_2 u_2 = 0, \tag{B}$$
$$a_1 u_1^2 + a_2 u_2^2 = \tfrac{2}{3}, \tag{C}$$
$$a_1 u_1^3 + a_2 u_2^3 = 0. \tag{D}$$

Ask yourself These equations are nonlinear. Is there a combination of them that factorises?

Take (D) minus u_2^2 times (B): the a_2 terms cancel and

$$a_1 u_1^3 - a_1 u_1 u_2^2 = a_1 u_1 (u_1 - u_2)(u_1 + u_2) = 0.$$

We rule out the first three factors in turn.

- If $a_1 = 0$: then (A) gives $a_2 = 2$, (B) gives $u_2 = 0$, and (C) reads $0 = \frac{2}{3}$. Impossible.
- If $u_1 = 0$: then (B) gives $a_2 u_2 = 0$, so $a_2 u_2^2 = 0$, and again (C) reads $0 = \frac{2}{3}$. Impossible.
- If $u_1 = u_2$: the two points coincide, (B) becomes $(a_1 + a_2)u_1 = 2u_1 = 0$, so $u_1 = u_2 = 0$ and (C) fails once more.

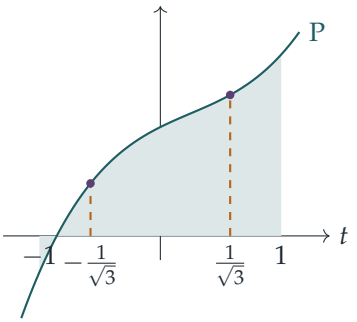
Hence $u_2 = -u_1$ with $u_1 \neq 0$. Now (B) becomes $(a_1 - a_2)u_1 = 0$, so $a_1 = a_2$, and (A) gives $a_1 = a_2 = 1$. Finally (C) gives $2u_1^2 = \frac{2}{3}$, so $u_1^2 = \frac{1}{3}$; (D) is then automatically satisfied. Therefore

$$\boxed{a_1 = a_2 = 1, \qquad u_1 = -\tfrac{1}{\sqrt{3}}, \quad u_2 = \tfrac{1}{\sqrt{3}}}$$

(or the same with u_1 and u_2 swapped), that is,

$$\int_{-1}^{1} \mathrm{P}(t)\,\mathrm{d}t = \mathrm{P}\Big(-\tfrac{1}{\sqrt{3}}\Big) + \mathrm{P}\Big(\tfrac{1}{\sqrt{3}}\Big) \quad \text{for every cubic P.}$$

The argument also shows that this is the *only* choice, apart from relabelling.



The shaded area equals the sum of the two dashed heights, whatever cubic is drawn.

Checks

- $\mathrm{P}(t) = t^2$: the integral is $\frac{2}{3}$ and the rule gives $\frac{1}{3} + \frac{1}{3}$.
- $\mathrm{P}(t) = 1 + t + t^2 + t^3$: the integral is $2 + \frac{2}{3} = \frac{8}{3}$; the rule gives $2 + 2 \cdot \frac{1}{3} = \frac{8}{3}$, since the odd terms cancel.
- The rule is *not* exact for t^4: it gives $2 \cdot \frac{1}{9} = \frac{2}{9}$, while the integral is $\frac{2}{5}$. So cubics really are the limit.
- Numerically, on 10^4 random cubics the two sides agreed to within 10^{-14}.

Common slips

- Assuming symmetry ($u_2 = -u_1$, $a_1 = a_2$) from the start. It gives the right answer quickly, but it doesn't show that no other choice works; the factorisation above does.

- Forgetting the cases $a_1 = 0$ or $u_1 = u_2$ when dividing through.
- Testing only $\mathrm{P}(t) = 1$ and t^2: the odd conditions (B) and (D) are what force the symmetry.

Beyond STEP

This is two-point *Gaussian quadrature*. With n well-chosen points and weights one can integrate every polynomial of degree up to $2n - 1$ exactly, twice what n equally spaced points manage. The best points are the zeros of the Legendre polynomials (here $\frac{1}{2}(3t^2 - 1)$, whose zeros are $\pm\frac{1}{\sqrt{3}}$), which are orthogonal on $[-1, 1]$. Orthogonal polynomials appear in first-year numerical analysis and in solving differential equations; Boyce and DiPrima, *Elementary Differential Equations*, meets Legendre's equation in its series-solutions chapter.

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